

GHOSTBRIDGE X NETMAKER

Netmaker is not a dependency. It is core code.

We built a zero-trust, compliance-grade AI control plane in Rust, and Netmaker is the spine: transport, identity root, and link zero of the evidence chain.

This is the first time anyone outside the project will have seen it. The meeting started as a trial/sales call, but the real point is simpler: we want a sharp Netmaker engineer to look closely and tell us what you think.

Rust control plane

Netmaker identity spine

No-NIC hardening pattern

OSCAL-gated actions

Confrontable AI

Suggested subject line: First human eyes: GhostBridge built on Netmaker

THE ONE PARAGRAPH

A mesh you can trust, not just connect

GhostBridge turns a Netmaker mesh into a trusted backbone. Traffic moves through decoy ingress, identity sled, Xray header injection, a per-hop interceptor, schema validation, OSCAL policy gates, and a tamper-evident Snowball ledger.

An AI layer reasons over verified history, but **it has no hands**: no execution rights, no direct control, and no ability to bypass the governed layer. It recommends into schema. A.N.N.A. Scribe and the Compliance Engine decide what is allowed to become real state.

WHY THIS IS DIFFERENT

■ AI you can cross-examine

Chatbot on top, Qdrant evidence on the bottom. Ask why something happened and the system answers with semantic receipts from the chain.

■ Chain is truth

Qdrant and Cozo are rebuildable projections. The canonical record is the hash-linked Snowball ledger, not the search database.

■ Model has no hands

The AI can reason, delegate, and recommend. It cannot directly execute. Hallucination becomes a rejected recommendation, not an incident.

■ One privacy dial

Same mesh and chain can run privacy-first or compliance-first. The identity-linkage layer changes, not the whole architecture.

GOVERNANCE STACK

Make the wrong thing impossible first

RUST

The compiler is the first enforcer: invalid code should not build.

SCHEMA

Invalid state is unrepresentable. In this system, plugin is schema; without validated schema, the object does not exist.

GEMMA + XRAY

Classify and route traffic. They direct; they do not authorize.

ZEROCLAW

Route to the right model by cost, reasoning depth, specialty, and policy.

A.N.N.A.

A.N.N.A. Scribe notarizes identity and Snowball session state. The Law Firm gates policy.

CHAIN

Every accepted action becomes a hash-linked, tamper-evident block.

THE LOOP

AI rebuilds semantic and graph projections from the chain, reasons over them, and produces the next governed recommendation.

NETMAKER HARDENING PATTERN

The part we want Netmaker to see

The reference pattern is a Netmaker control plane with no exposed service port: loopback-only services, Incus proxy devices, Unix sockets under **/run/netmaker**, and Xray routing by SNI behind WireGuard identity.

■ No NIC posture

netmaker, **netmaker-mq**, and **netmaker-ui** are treated as loopback/socket services, not public daemons.

■ Sockets not ports

The attack surface moves to socket files such as **/run/netmaker/api.sock** and **ui.sock**, behind the identity gate.

■ Xray routing

wg-xray terminates and routes traffic by domain to the correct internal socket-backed service.

WHY NETMAKER IS CORE

You are at the root of the identity story

- Netmaker is represented as a first-class state plugin, not a bolt-on integration.
- WireGuard peers are the identity root: **session_id = Argon2(PSK, salt = WG_pubkey)**.
- Remove Netmaker and the identity layer loses link zero of the evidence chain.
- GhostBridge turns hardened Netmaker deployments into a regulated-room story: privacy, compliance, accountability, and AI evidence.

WHAT WE WANT

Not a pitch. A first look.

We are not asking for a purchase decision, a partnership commitment, or homework. We want a knowledgeable human at Netmaker to see what has been built on top of your layer and react like an engineer.

The architecture is implemented far enough to deserve that reaction. Some edges are still being hardened, but the core idea is real and implementable with what exists today.

Netmaker connects the machines. GhostBridge proves who they are and that policy was enforced, every hop, auditor-grade, in an AI loop you can cross-examine.

Confidential pre-meeting brief. Self-contained HTML; no external assets, no tracking, no remote dependencies.